

backend/api/graph.py Documentation (Roman Urdu)

Yeh API file application ke "Brain Map" (Knowledge Graph) k liy ha, jis k ander baton aur files k reference apas mb jurre huwey hn nodes list mein. Sath e Web scraping k controls b ism e hyn.

File Ka Maqsad (Purpose)

Vector dbs srf meaning mapping hy lkn Knowledge graph relationships search krta h (like: X related ha Y s, Y connected h Z pr), ye script us graph k liye information expose kartab Frontend pr (NetworkX / D3.js ky liye).

Code Blocks aur API Endpoints ki Tafseel

1. GET /visualization

- **Kya karta hai?:** Node-Link format JSON array bna k UI ma bhejta jis say Frontend D3.js istimaal kar ky aik khoubsoorat jala-numma (force-directed) 3D design draw karta ha k system me kya data stored ha.

2. Graph Search & Delete (/search, /entity/{id}, /delete)

- **Kya karta hai?:** Database ky vertices mai text dhoondna ar agar koi memory line aesi ha jo user urana chhhta ha tu backend use graph m se flush kr deta hy.

3. GET /analysis

- **Kya karta hai?:** Yeh Node Analysis run krti ha. Jse Google PageRank karta hy, waehay y chiz check krti h k poore knowledge memory mb konsa point ya file sab se aham aur connected ha (Central node) taki summary bnanay waqtb sb sy phely ussey prioritiz karyn.

4. Web Scraping (/scrape , /scrape/preview)

- **Kya karta hai?:** User web link dale ar UI kaey Scrape kard, tou yeh system file phelay config option dkhata ha eejazat hb ? Pher `web_scraper.scrape_and_ingest()` path par chla dtb tbky sari website scrap ho jaey aur graph m inject houjeye as Knowledge! Preview

wala path bs scrap dikhta ha lkn save nai krwata data mein.